

PROBLEM STATEMENT TITLE:

Post-Harvest Preservation, Water Efficiency & Logistics Visibility

THEME: SUSTAINABLE AGRICULTURE

PRESENTED BY
SHADOW SYNDICATES

TEAM MEMBERS

VISHNUVARDHAN B
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Every harvest has a race against time to preserve its freshness.

REAL WORLD PROBLEM

Post Harvest Losses

Significant fresh produce is lost before reaching consumers due to heat, poor storage, and handling.

Limited Support for Smallholders

FPOs pool produce from many smallholders, but often run cooling with no data on effectiveness or cost.

Changing Environmental Conditions

Evaporative cooling effectiveness depends on temperature and humidity, and can shift as conditions change.

Water and Energy Used for Cooling

Every cooling cycle uses water and electricity, often without operators knowing if it's worth the resource use.



PROBLEM STATEMENT:

What Does This Problem Aim to Solve?

- Makes cooling effectiveness visible, not assumed
- Confirms produce is actually cooling, not just that the cooler is running
- Flags when cooling stops working due to humidity
- Tracks water and energy used per cooling cycle
- Creates a batch-level record for every cooling cycle
- Gives FPOs data to support better cooling decisions

Why We Chose This Problem?

- Directly matches Challenge 01's call for low-cost sensing and post-harvest loss reduction
- Field heat removal is a real, well-documented gap in India's post-harvest chain
- FPOs are the realistic, scalable entry point — not individual farmers
- Cooling exists, but effectiveness and resource use stay invisible

Who Faces This Issue?



PROPOSED SOLUTION

AgroTrace Cool — Smart Monitoring for Farm-Gate Cooling

Harvested produce carries **field heat** and **starts losing quality within hours**. **AgroTrace Cool** is a low-cost **evaporative cooling chamber** built with a sensor-driven monitoring system — confirming whether the produce is actually cooling, flagging when **effectiveness drops**, and **recording** every cooling cycle for the FPO.



KEY FEATURES:

- **Produce-Temperature Monitoring** — Direct temperature sensing on the produce, for **accurate cooling** decisions
- **Adaptive Cooling Logic** — Uses temperature and humidity to decide **COOL, REDUCE, or STOP**
- **Resource Tracking** — Logs **water and energy** used in every cooling cycle
- **Android App For Batch Records** — Displays live status, cooling graphs, and cycle history for the FPO

WHAT MAKES IT DIFFERENT?

BEFORE — Basic Cooling Setups



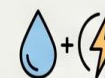
Cooler runs blindly, regardless of conditions



No idea if produce is actually cooling



Keeps running even when humidity makes cooling ineffective



No record of water or energy used



“Feels cool” is the only check available



No record after the cycle ends



No data to show funders or buyers

AFTER — With AgroTrace Cool



Checks temperature & humidity before cooling



Monitors produce's own temperature directly



Detects low effectiveness and reduces/stops



Logs water & energy per cooling cycle



Clear status: **COOL / REDUCE / STOP / TARGET REACHED**



Batch history stored in the Android app



Structured records ready for reporting

IMPACT & BENEFITS



What's the Current Gap in the System?

- Cooling runs without checking if it's actually working
- No data on water or energy used per cycle
- No record after the cycle ends
- Operators rely on guesswork, not readings



How Does AgroTrace Cool Fill That Gap?

- ✓ Monitors produce temperature directly, not just chamber conditions
- ✓ Flags when humidity makes cooling ineffective
- ✓ Logs water, energy, and duration for every cycle
- ✓ Stores batch history in the Android app for FPO review

Without Cooling
(Stored at normal temperature)

- ✗ Gets heated quickly
- ✗ Dries out
- ✗ Loses freshness
- ✗ Shorter shelf life

With AGROTRACE COOL
(Controlled cooling & monitoring)

- ✓ Stays fresh longer
- ✓ Maintains moisture
- ✓ Better quality
- ✓ Extended shelf life

REAL-WORLD IMPACT

Making the cooling process measurable at the FPO level



Cooling Visibility

Operators know whether cooling is working, not guessing



Resource Tracking

Water and energy use becomes visible and trackable

SENSE → UNDERSTAND → ACT → RECORD



Better Records

FPOs gain structured records for internal decisions and reporting



Less Unnecessary Cooling

Reduces resource waste from cooling that isn't helping

TECHNICAL APPROACH



Hardware + Software powering measurable farm-gate pre-cooling

HARDWARE

01 CORE SENSING



DHT22
(Temp. & Humidity)



DS18B20
(Produce Temperature)



ESP32
(Main Controller)

Measures ambient temperature, relative humidity and representative produce-core temperature.

02 COOLING HARDWARE



DC Fan
(Airflow)



DC Water Pump
(Water Supply)



Wet Jute / Coir Medium
(Evaporative Pad)

Creates airflow and evaporative cooling through the wet medium.

03 CONTROL & STATUS



Relay / MOSFET Driver
(Switches Fan & Pump)



OLED Display
(Local Status)



LED / Buzzer
(Status Indication)

Safely switches the fan and pump and provides immediate local operating status.

04 POWER & BUILD



Power Bank
(5V USB)



Enclosure
(Weather Protection)



Tubing
(Water Flow)



Solar Panel
(Future Scope)



Wires & Connectors
(System Build)

Low-voltage power and practical construction for the system (solar option for future scope).

SOFTWARE

05 ESP32 FIRMWARE



- C / C++
- Arduino IDE
- Sensor Libraries
- Decision Logic
- Local Logging

Reads sensors, evaluates cooling conditions, controls, controls and records cooling cycles.

06 ANDROID APPLICATION



- Kotlin
- Android Studio
- Jetpack Compose
- Operator Dashboard
- Batch Logistics
- Cooling Graphs

Displays live cooling status, produce temperature, batch history and cycle data.

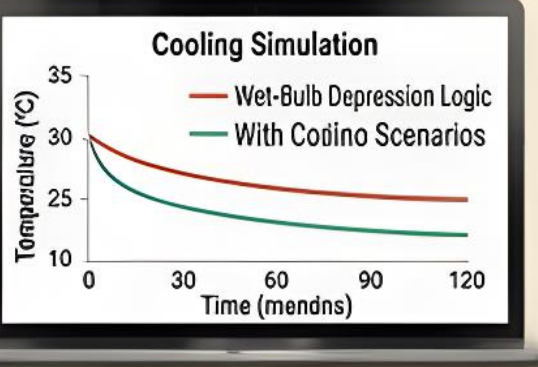
07 DATA SYNC & STORAGE



- Bluetooth Low Energy (BLE)
- SQLite (Local Storage)
- CSV Export
- Batch-wise Records

Transfers cooling data from the ESP32 to the Android application and stores structured batch-level records.

08 COOLING SIMULATION



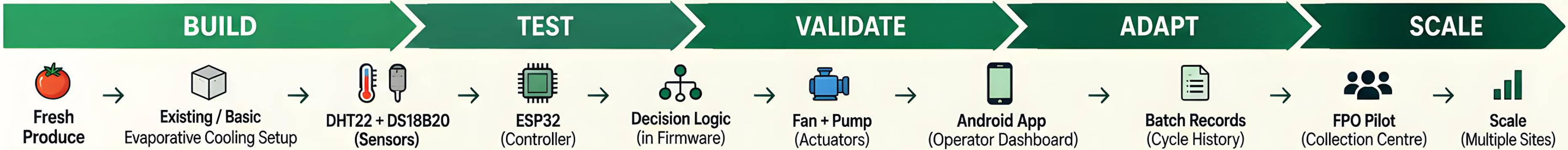
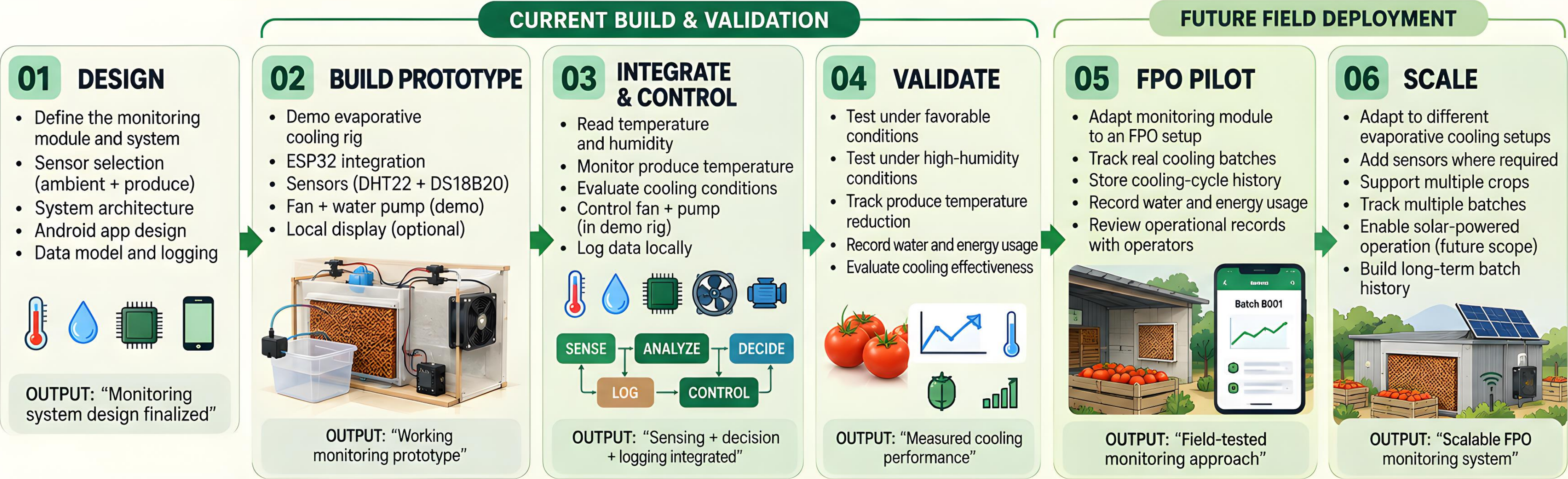
- Python
- Wet-Bulb Depression Logic
- Stage-1 Demo Scenarios
- Temperature-vs-Time Modelling

Simulates favorable and unfavorable cooling conditions for the Stage-1 demonstration.



IMPLEMENTATION ROADMAP

From working prototype to FPO-ready monitoring

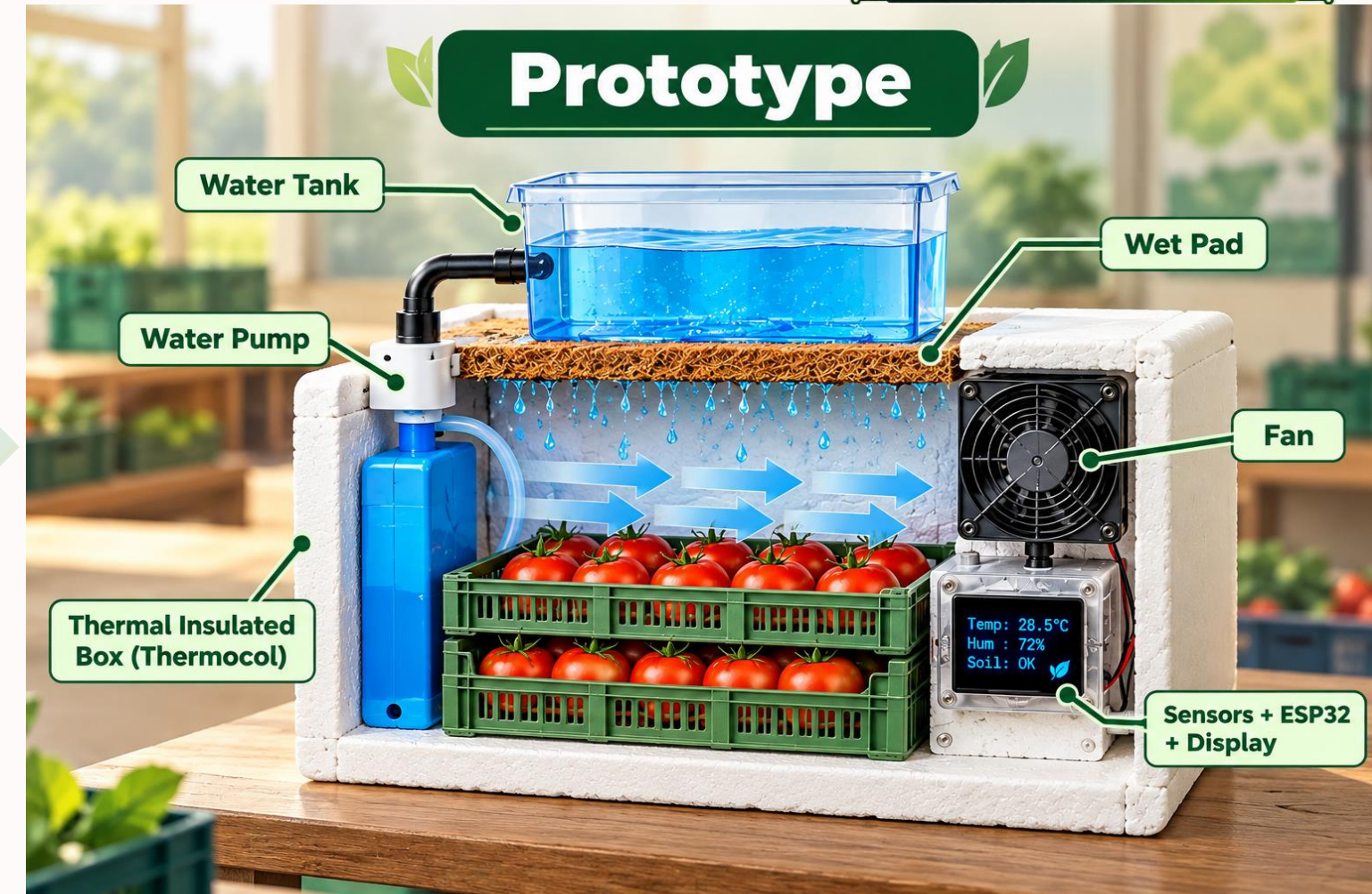


BUILD THE DEMO → MEASURE THE PROCESS → VALIDATE THE RESULTS → ADAPT FOR FPOs → SCALE

FEASIBILITY & VIABILITY

Prototype Feasibility

- ✓ Low-cost, readily available components
- ✓ Simple cooling chamber + sensor setup
- ✓ ESP32 handles sensing, control and logging
- ✓ Buildable and testable at prototype scale



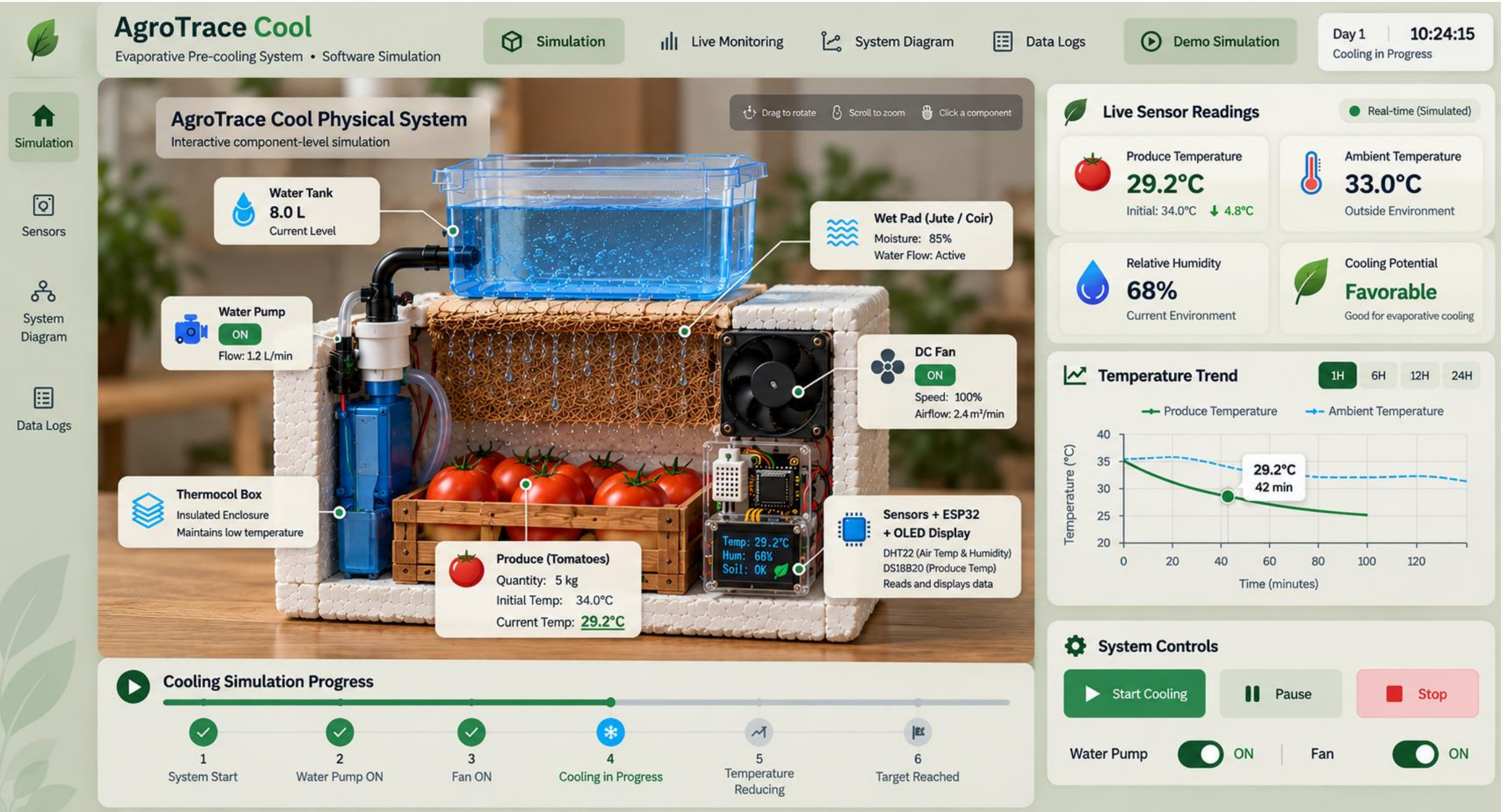
FPO Adaptable

- Monitoring approach designed for FPO cooling operations
- Can be adapted to larger cooling chambers
- Batch-wise temperature, water and energy records
- Adds monitoring without replacing the cooling method

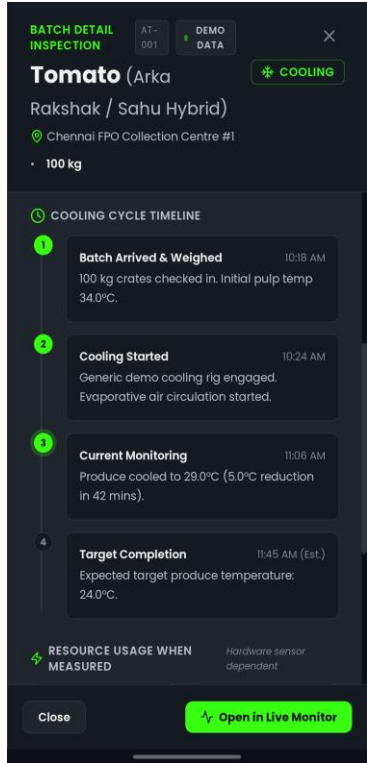
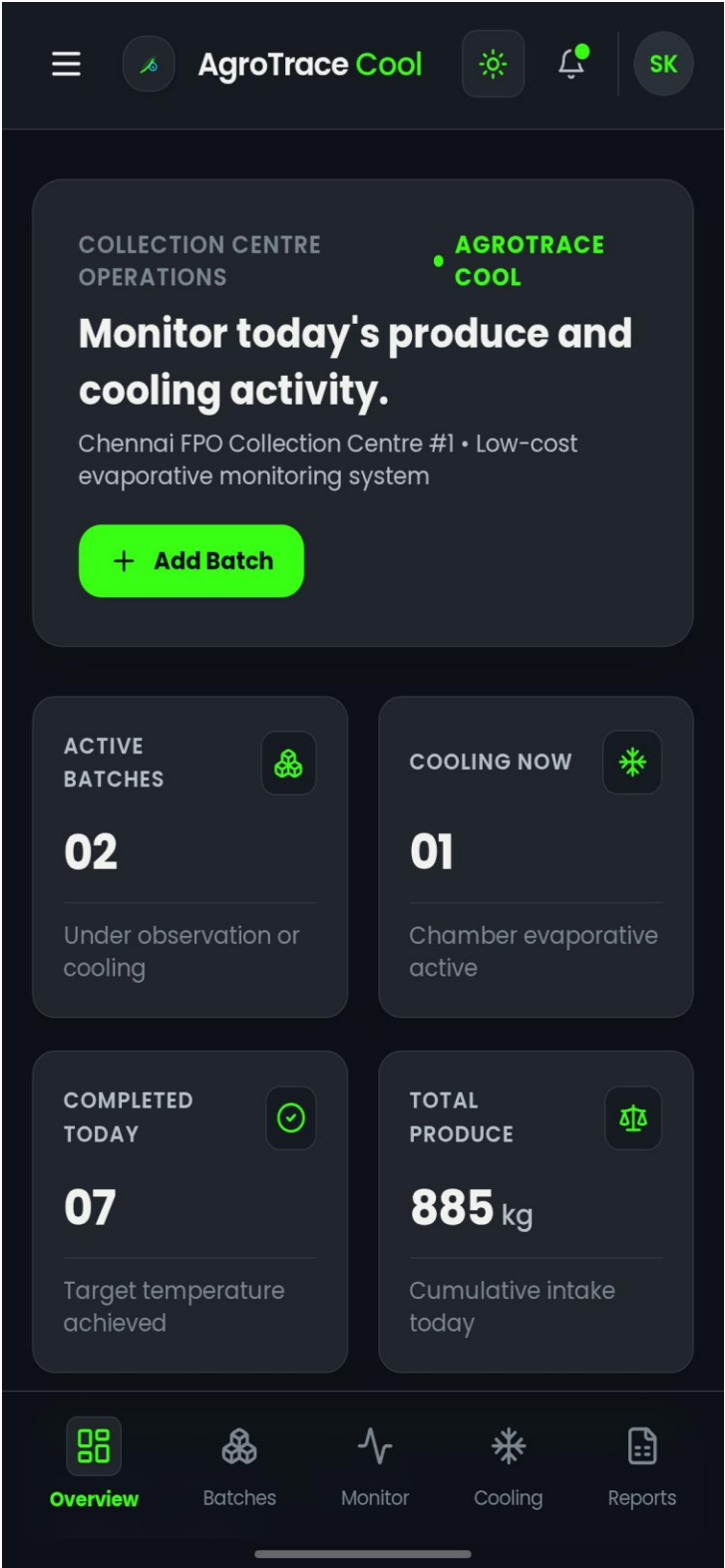
Scalable & Sustainable

- ✓ Prototype → FPO pilot → larger deployment
- ✓ Modular sensors and monitoring units
- ✓ Resource use measured before scale-up
- ✓ Grid, battery or solar power can be considered for deployment

AgroTrace Cool — Monitoring System Prototype



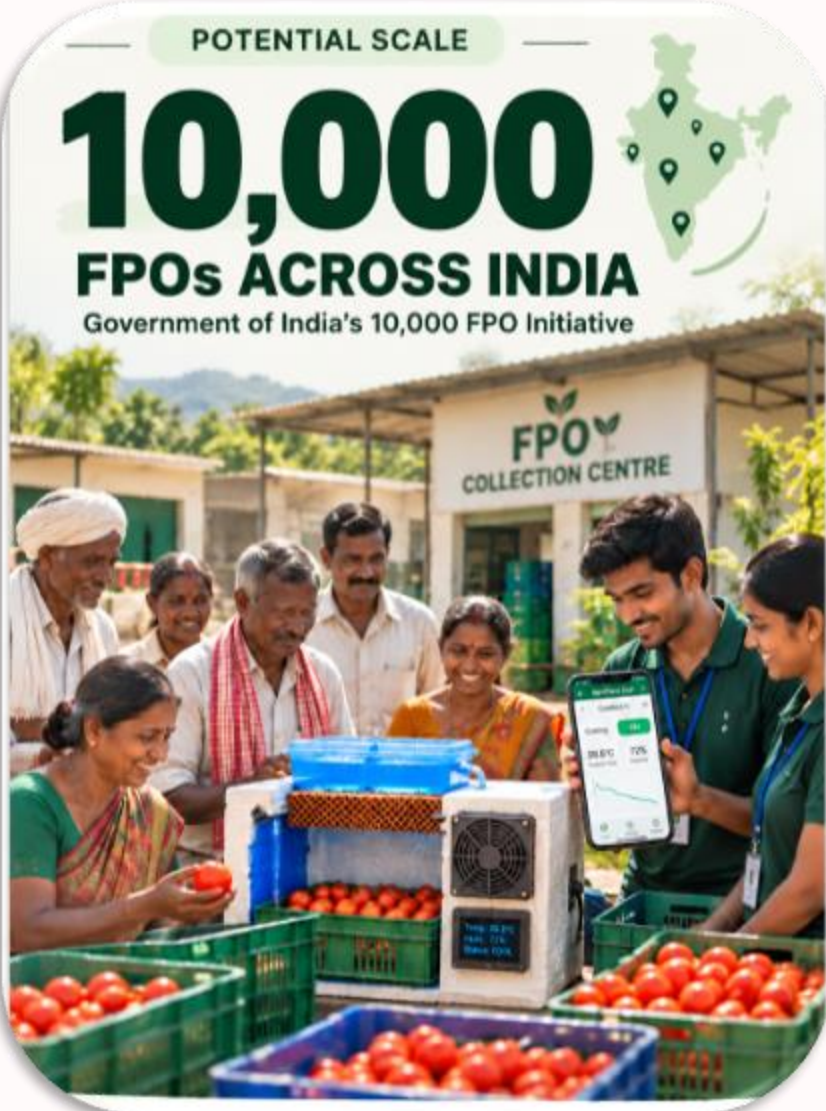
Monitoring Website Demo



RESEARCH AND REFERENCES

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- **Smart Indigenous Cooling Chamber (2024–2026)** — Govt.-funded IoT evaporative cooling research, Telangana <https://www.indiascienceandtechnology.gov.in/hi/node/170680>
- **CoolCrop** — Farm-level, FPO-aggregation cooling chambers (NITI Aayog) <https://frontiertech.niti.gov.in/story/reimagining-cold-chains-tech-driven-storage-solutions-for-indias-perishable-agriculture/>
- **FreshBox** — Dual-stage evaporative cooling for Indian post-harvest loss (Elephant Design) <https://www.elephantdesign.com/perspective/innovation-mint/freshbox>
- **Forced-Air Evaporative Cooling Chamber** — IoT-connected pilot near Bhuj, India (MIT D-Lab / CoolVeg) <https://jwafs.mit.edu/projects/2021/mobile-evaporative-cooling-rooms-vegetable-preservation>
- **ICCVAI Scheme under PMKSY** — Govt. cold-chain infrastructure funding (Ministry of Food Processing Industries) <https://static.pib.gov.in/WriteReadData/specificdocs/documents/2025/nov/doc2025114684601.pdf>
- **FPO-Led Shared Cold Storage Models** — Andhra Pradesh, Karnataka case studies (SELCO Foundation) <https://india.mongabay.com/2025/11/solar-powered-cold-storage-empowers-smallholder-farmers/>



OUR PROJECT REPOSITORY:

[GITHUB LINK](#)

- ✓ Research Papers & References
- ✓ Prototype Simulation & Testing
- ✓ Technical Documentation
- ✓ Future Scope



MEET THE TEAM:



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